Force

In physics, a **force** is a push or pull acting upon an object as a result of the object's interaction with another object.

It is an external agent capable of altering an object's state of rest or motion, causing it to accelerate, decelerate, or change direction.

Key Characteristics

- **Vector Quantity** – Forces have both magnitude (strength) and direction.

- **Standard Unit** – The metric unit for force is the **Newton (N)**.

One Newton is the force required to accelerate a 1 kg mass at a rate of **1 m/s^2**.

- **Formula** – Newton’s Second Law of Motion:

\[

F = m a

\]

where

* \(F\) = force,

* \(m\) = mass,

* \(a\) = acceleration.

Common Types of Forces

- **Contact Forces** – Applied through physical contact between objects (e.g., friction, tension, air resistance, or a physical push/pull).

(Note: Other force categories such as non-contact forces exist, but are not covered in the provided OCR text.)